Banking Loan Risk Analysis Application - System Architecture

End-to-End Microservice Architecture: ETL, Relational Data Mart, Risk Rules Engine & Analytics Dashboard

1. Data Ingestion

- UCI Credit Card Clients Data
- 30,000 Historical Records
- Demographics (Age, Sex, Edu)
- 6-Mo Bills & Payment History
- Default Payment Indicators

2. ETL & Quality Service

- Pandas Data Extraction (XLS/CSV)
- Data Cleansing & Field Decoding
- Age Group & Ratio Engineering
- Data Quality Audit (8 Rules)
- Record Reconciliation & MD5 Hash

3. Relational Storage

- Database: MySQL / SQLite
- customers (Demographics)
- credit_profiles (Risk & Limits)
- bills & payments (Time-series)
- segment_kpi_summaries (Indexed)

4. Risk & Rules Engine

- Multi-factor Risk Scoring (0-100)
- Low / Medium / High Risk Bands
- Underwriter Review Triggers
- Utilization & Repayment Ratios
- What-If Applicant Simulator

5. FastAPI REST API Layer (Backend)

- Uvicorn ASGI Microservice Architecture
- /auth: JWT RBAC (Analyst vs Administrator)
- /customers: Paginated CRUD, Search, Filter & CSV Export
- /risk: Risk Scoring & Manual Review Queue Workflow
- /analytics: Portfolio KPIs, Demographic Segments & Trends
- /etl: Incremental Pipeline Trigger & Audit Logs

6. Streamlit Interactive Dashboard (UI)

- Executive Portfolio KPIs & Risk Donut Breakdown
- Customer Explorer & 6-Month Ledger Drilldown
- Underwriter Manual Review Approval/Rejection Queue
- Demographic Drilldowns & Cashflow Trends (Plotly)
- Interactive What-If Loan Underwriting Simulator
- ETL Control Center & SQL Analytics Studio